MATHEMATICS CLASS – X WBBSE 2025

1. Choose the correct option in each case from the following questions: (1X6=6)

i. Simple Interest of Rs. Y for Z months at the rate of X% per annum is

(a) Rs. $\frac{XYZ}{1200}$ (b) Rs. $\frac{XYZ}{100}$ (c) Rs. $\frac{XYZ}{200}$ (d) Rs. $\frac{XYZ}{120}$

ii. If a:2 = b:5 then how many % of b is a

(a) 20 (b) 30 (c) 40 (d) 50

iii. AB is the diameter of a circle with centre O. If the chord AC subtends angle of 60^0 at centre, then value of $\angle OCB$ is

(a) 20^0 (b) 30^0 (c) 40^0 (d) 50^0

iv. If lengths of three sides of a triangle are $sec\theta$, 1, $tan\theta$ $(\theta \neq 90^0)$. The value of greatest angle of the triangle is

(a) 30^0 (b) 45^0 (c) 60^0 (d) 90^0

v. The radius of a right circular cylinder and a hemisphere is equal and their volumes are also equal. The height of the hemisphere is greater than the height of the cylinder by

(a) 25% (b) 50% (c) 100% (d) 200%

vi. 27, 31, 46, 52, x, y+2, 71, 79, 85, 90 are in increasing order. If the median is 64 then the value of x+y is

(a) 125 (b) 126 (c) 127 (d) 128

2. Fill in the blanks (any Five):- (1X5=5)

i. In a business Pintu invests $1\frac{1}{2}$ times of principla amount of Aman and David invests $2\frac{1}{2}$ times of principla amount of Aman . The ratio of Principal amount of Aman, Pintu and David is ____________

ii. If $x(4-\sqrt{3}) = y(4+\sqrt{3}) = 1$, then value of $x^2 + y^2 =$____________

iii. If two circles in a plane have three common tangents, then they will touch each other ____________

iv. If $sin^2 0 + 2xcos^2 0 = 1$, then value of x is____________

v. The greatest volume of a solid cone which can be cut off from a solid hemisphere of radius r unit is ____________

vi. *x* is mean of *(p+q)* numbers. If *y* be the mean of *p* numbers, then mean of remaining *q* number is ____________

3. Write True or False (any Five):- (1X5=5)

i. In a joint business one of the two friends invests Rs *xyz* for *y* months and the other Rs. y^2z for *x* months. At the end of the agreement, the profit would be distributed in the ratio of *x : y*.

ii. If the sum of the square of the roots of the equation $6x^2 + x + k = 0$ is $\frac{25}{36}$, then the value of *k* is 12.

iii. ABCD is a cyclic quadrilateral. If $\angle ADB = x^0$, $\angle ABD = y^0$, then $\angle BCD$ is $(x + y)^0$.

iv. If $0^0 < \theta < 90^0$, then $sin\theta < sin^2\theta$.

v. If x be the volume, y be the area of the base and z be the height of a cone then the value of $\frac{x}{yz} = 3$.

vi. $Mode = 2 \times Median - 3 \times Mean$.

4. Answer the following questions (any ten):- (2X10=20)

i. If the annual rate of simple interest decreased from 5.5% to 4.5%, then the total interest is decreased by Rs. 250. Find the capital.

ii. The capital of A and B in a business are in the ratio of 3 : 2. After donating 5% of the profit B receives Rs 798. What is the total profit ?

iii. If $\frac{x}{2} = \frac{y}{3} = \frac{z}{4}$, then find the value of $\frac{3x+4y+8z}{x+3y}$.

iv. If $x \propto \sqrt{y}$ and $y = a^2$, when $x = 2a$, then find the value of $x^2 : y$.

v. Perimeters of two similar triangles are 27 cm and 16 cm. If the length of a side of the first triangle is 9 cm, then find the length of corresponding side of the second triangle.

vi. From an external point P of a circle with centre O, two tangents PS and PT are drawn. QS is a chord of the circle parallel to PT. If $\angle SPT = 80^0$, then find the value of $\angle QST$.

vii. O is a point inside a rectangle ABCD such that OB = 6 cm, OD = 8 cm and OA = 5 cm. Find OC.

viii. If $sin\ (\theta + 30^0) = cos\ 15^0$, then find the value of $cos\ 2\theta$.

ix. If $cos^4\theta - sin^4\theta = \frac{2}{3}$, then find the value of $1 - 2sin^2\theta$.

x. If x be the number of edges of a parallelepiped and y be the number of surfaces then for what least value of a, $(x + y + a)$ is a perfect square.

xi. The ratio of length of radius of wo solid right circular cylinders is 2 : 3 and the ratio of their heights is 5 : 3. Find the ratio of their area of curved surfaces.

xii. The median of first $(2n + 1)$ natural numbers is $\frac{n+103}{3}$. Find the value of n.

5. Answer any one question:- (1X5=5)

i. In a joint business the capital of Samar and Mohim are Rs 20,000 each. After 6 months Samar invests Rs 5000 more, but Mohim withdraws Rs. 5000. At the end of the year if the profit is Rs. 32000, then find the share of profit of Samar and Mohim.

ii. Divide Rs 21866 into two parts such that the amount of the first part for 3 years and amount of the second part for 5 years are equal at the same rate of 5% compound interest.

6. Answer any one question:- (1X3=3)

i. Divide 16 into two parts in such a way that twice the square of the greater part is 164 more than the square of the smaller part.

ii. Solve: $\frac{x+3}{x-3} + \frac{x-3}{x+3} = 2\frac{1}{2}$, $(x \neq -3, 3)$.

7. Answer any one question:- (1X3=3)

i. If $\left(x^3 - \frac{1}{y^3}\right) \propto \left(x^3 + \frac{1}{y^3}\right)$, then show that $x \propto \frac{1}{y}$.

ii. If $x = \frac{4\sqrt{15}}{\sqrt{5}+\sqrt{3}}$, then find the value of $\frac{x+\sqrt{20}}{x-\sqrt{20}} + \frac{x+\sqrt{12}}{x-\sqrt{12}}$.

8. Answer any one question:- (1X3=3)

i. If $(b + c - a)x = (c + b - a)y = (x + b - c)z = 2$, then prove that

$$\left(\frac{1}{x} + \frac{1}{x}\right)\left(\frac{1}{y} + \frac{1}{z}\right)\left(\frac{1}{z} + \frac{1}{x}\right) = abc$$

ii. If $\frac{x}{y} = \frac{a+2}{a-2}$, then find the value of $\frac{x^2-y^2}{x^2+y^2}$.

9. Answer any one question:- (1X5=5)

i. Prove that in a cyclic quadrilateral opposite angles are supplementary.

ii. State and prove Pythagoras theorem.

10. Answer any one question:- (1X3=3)

i. AB is the diameter of the circle with centre O. From a point P on the circle, a perpendicular PN is drawn on AB. Prove geometrically that: $PB^2 = AB \times BN$

ii. O is the circumcenter of the triangle ABC, $OD \perp BC$. Prove that: $\angle BOD = \angle BAC$

11. Answer any one question:- (1X5=5)

i. Find the value of $2\sqrt{3}$ geometrically.

ii. Draw a triangle whose sides are of lengths 6cm, 8 cm and 10 cm. Draw the incircle of this triangle.

12. Answer any two questions:- (3X2=6)

i. If $\sin x = m \sin y$ and $\tan x = n \tan y$, then show that $\cos^2 x = \frac{m^2-1}{n^2-1}$.

ii. If $\tan\theta = \frac{5}{7}$, then find the value of $\frac{5\sin\theta+7\cos\theta}{7\sin\theta+5\cos\theta}$

iii. Two unequal arcs of ratio 5 : 2 of a circle subtend angles at the centre. If the second angle is 30^0. Find the first angle in circular measure.

13. Answer any one question:- (1X5=5)

i. From the middle of a ground Habu observers a flying bird towards north at an angle of elevation 30^0, after 2 minutes he observes the same bird at an angle of elevation 60^0 towards south. If the bird always flies in a straight line at a height $50\sqrt{3}$ meter from the ground level, find its velocity.

ii. Distance between two pillars is 150 m, height of one pillar is three times the other pillar. From the mid point of the line joining their foot the angle of elevation of the top of the two pillars are complementary. Find the height of the smaller pillar.

14. Answer any two questions:- **(4X2=8)**

i. Curved surface area of a right circular cone is $154\sqrt{2}$ sq cm and radius of the base 7 cm. Find its vertical angle.

ii. The height of a right circular cylinder is twice the radius of the base. If the height is 6 times of the radius, then the volume of the cylinder is 539 cubic decimeter more than the previous volume. Find the height of the cylinder.

iii. Melting a solid lead sphere of diameter 12 cm, three small solid spheres are made. If the ratio of the diameter of the small spheres is 3 : 4 : 5, then find the radius of each small spheres..

15. Answer any two questions:- **(4X2=8)**

i. The age of 100 persons present in a workshop is given in the following table. Find the average age of 100 people by any method.

Age (years)	10-20	20-23	30-40	40-50	50-60	60-70
No. of People	8	12	20	22	18	20

ii. The median of the following data is 32. Find the value of x and y if x+y=100.

Class Interval	0-10	10-20	20-23	30-40	40-50	50-60
Frequency	10	x	25	30	y	10

iii. Draw an ogive on the graph paper after calculating the cumulative frequency (less than type) of the following data:

Class Interval	0-10	10-20	20-23	30-40	40-50	50-60	60-70
Frequency	1	6	15	20	15	6	1

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